Week 8 Mini-Project

External Data Medallion Pipeline and Power BI

Team project | Open/external data | Python cleaning | Bronze/Silver/Gold | Gold data model | Power BI | Git | Scrum

Mini-project overview

In this mini-project, your team will choose realistic external datasets and build a small but coherent data engineering solution. The goal is to ingest, clean, validate, model and report data using a clear Medallion architecture: Bronze, Silver and Gold. Your final Power BI report must use the Gold layer as its source.

The project is intentionally open in business topic, but fixed in engineering shape. Your team may choose the subject area, but the solution must include at least two joinable data sources, Python-based ingestion/cleaning work, a Gold data model, validation checks, documentation, Git usage and Scrum-style project management.

Team setup

- Work in a team and use Scrum-style project management: create a Product Backlog, choose tasks, track status and review progress regularly.
- Use Git as the code repository for notebooks, Python scripts, SQL scripts, documentation, source descriptions and Power BI-related files or screenshots.
- Agree on team responsibilities. Responsibilities may rotate, but every team member must have real implementation work, not only documentation or presentation work.
- Every team member must participate in Python data ingestion, cleaning, validation or transformation work. The team must be able to show who worked on which Python cleaning task.
- Use pair work where useful, but avoid one person doing all Python work while others only observe.

Important participation requirement

- Your Product Backlog must include Python cleaning or validation tasks assigned to every team member or pair.
- During instructor review, be ready to show which notebook cells, scripts, commits or pull requests each team member contributed to.
- A good division is: source A profiling/cleaning, source B profiling/cleaning, join/key standardization, validation/rejected rows, Gold model preparation, Power BI and final presentation.

Business scenario

Your team chooses a business or analytical scenario based on public, open or otherwise approved data. The scenario must be realistic enough that the datasets can be joined and modeled into a useful Gold layer.

Examples of suitable questions: How does weather relate to public transport usage? How do energy prices vary by region and time? How do sports results connect to team or country metadata? How can open statistics be enriched with population or region data?

Learning goals

- Select realistic, joinable external datasets and describe their grain, keys and limitations.
- Design a Bronze/Silver/Gold Medallion architecture that has real quality differences between layers.
- Use Python and Pandas to ingest, profile, clean, standardize and validate data.
- Create a Silver layer with cleaned and typed data, including review/rejected rows where appropriate.
- Dbt can be used but is not a must. Airflow usage is not expected.
- Design a final Gold data model that supports reporting and analysis.
- Use SQL where useful for staging, validation, table creation, querying or Gold-model support.
- Create Power BI reports from the Gold layer, not directly from raw source data.
- Use Git and Scrum practices to manage a small data engineering project.
- Document your architecture, pipeline and model.
- Present the architecture, data model, pipeline, quality checks and report results clearly.

Required project parts

Part 1 - Dataset selection and problem statement

- Choose at least two data sources. At least one source must come from the web, an open-data portal, an API, or a downloadable public dataset.
- The sources must be joinable or enrichable. There must be a meaningful relationship between them, for example location, time, product, team, region, category or country.
- Write a short problem statement or analysis goal: what business or analytical questions should the final Gold layer and Power BI report answer?

Part 2 - Product Backlog and project planning

- Create a Product Backlog with clear Product Backlog Items (PBIs).
- Include PBIs for dataset selection, source profiling, Python ingestion, Bronze storage, Silver cleaning, validation checks, Gold data model, Power BI, Git/review work and final presentation.
- Each PBI should have an owner or responsible pair and a current status.
- Keep the scope realistic. It is better to finish a smaller working pipeline than to design a large unfinished solution.

Part 3 - Medallion architecture plan

- Create a Medallion architecture plan before implementing the full pipeline.
- Bronze must represent raw or near-raw ingested data. Store the original structure as much as practical and keep source metadata such as source name, file/API name, load timestamp or run id.
- Silver must contain cleaned, standardized, typed and validated data. This is where column names, dates, numeric values, category names and join keys should become reliable.
- Gold must contain the final reporting-ready data model. The Gold layer should answer the selected business questions and be suitable for Power BI.
- The plan must show source -> Bronze -> Silver -> Gold -> Power BI. It can be a diagram, table or clear written plan.

Part 4 - Python ingestion and cleaning

- Use Python and Pandas (or Polars) for ingestion, profiling, cleaning, standardization and validation work.
- Read data from files, web downloads, APIs or prepared extracts depending on your selected sources.

- Clean and standardize columns such as dates, numeric values, identifiers, category names, countries, regions, locations or status values as needed.
- Create validation outputs such as rejected rows, review rows, validation summaries or data quality metrics.
- All team members must participate in Python cleaning or validation work. This is a project requirement, not an optional nice-to-have.

Part 5 - Silver data and data quality checks

- Create Silver outputs that are clearly cleaner and more usable than Bronze outputs.
- Include meaningful data quality checks. Examples: not-null keys, duplicate keys, invalid dates, invalid numeric values, unmatched lookup keys, impossible values or row count checks.
- Document what happens to invalid rows: rejected output, review output, correction logic or documented exclusion.

Part 6 - Final Gold data model

- Create a final Gold data model. This can be a simple star or snowflake schema
- Write down the grain of each main Gold fact or mart table: what does one row represent?
- Use dimension-like lookup tables where useful: date, location, region, product, team, country, station, category or similar. Depending on data this can be simple one.
- Keep source/business keys as useful normal columns. If you create surrogate keys, document how they are generated.

Part 7 - SQL usage

- SQL may be used for table creation, loading to a database, validation queries, joins, aggregations or Gold-model queries.
- SQL is allowed and can be used heavily, but it does not replace the required Python ingestion and cleaning work.
- Document the database objects and how they are created or refreshed.

Part 8 - Power BI reports from Gold

- Create Power BI reports from Gold outputs only. Do not connect Power BI directly to the raw Bronze data as the main reporting source.
- Create at least two useful report pages or views.
- Include relevant measures such as counts, totals, averages, rates, durations, rankings, trends or comparisons depending on your selected topic.
- Make the report understandable for a business user. Use clear titles, filters/slicers and readable visuals.

Part 9 - Git and Scrum evidence

- Use Git for version control. The repository should contain code, notebooks, SQL, source descriptions, architecture notes and project documentation.
- Commit regularly with meaningful messages. Pair commits are acceptable, but the team must be able to explain who did what.
- Use Scrum-style workflow: Product Backlog, task ownership, daily/regular status updates and a Definition of Done.

Part 10 - Runbook and final presentation

- Create a short runbook that explains how to run the project from source data to Gold outputs and Power BI refresh. Runbook can be a document.

- Prepare a final presentation that shows the selected datasets, Medallion architecture, Python cleaning, validation checks, Gold data model and Power BI report.
- Mention limitations and what you would improve next if there was more time.

Suggested data source patterns

Pattern	Why it works
Weather + city/country indicators	Good join keys and time/location dimensions.
Public transport + weather	Operational data can be enriched by external conditions.
Energy prices + weather	Time-series joins, date handling and aggregation.
Sports results + team/country metadata	Facts, dimensions, rankings and season logic.
Open statistics + population/region data	Granularity mismatch and enrichment challenge.
Events + location metadata	Easy Gold model for counts, trends and regional comparisons.

Minimum deliverables

Deliverable	Expected contents
Selected datasets and source description	Dataset names, links or file names, source type, grain, important columns, expected join keys, limitations and fallback plan if a web/API source fails.
Product Backlog	PBIs for source selection, ingestion, Bronze, Silver, cleaning, validation, Gold model, Power BI, Git/Scrum, presentation and current status.
Medallion architecture plan	Diagram or clear plan showing sources -> Bronze -> Silver -> Gold -> Power BI, including what changes between layers.
Python ingestion and cleaning work	Notebooks or scripts that read, profile, clean, standardize and validate data. Must include evidence that all team members participated.
Bronze outputs	Raw or near-raw ingested data with source metadata.
Silver outputs	Cleaned, typed, standardized and validated data plus rejected/review outputs if needed.
Final Gold data model	Star schema, dimensional model or reporting mart. Include fact/mart grain and key columns.
Validation checks	Queries or Python checks that prove row counts, key completeness, date validity, duplicate handling and important totals are reasonable.
Power BI report from Gold	Power BI file or screenshots plus short explanation of report pages and business questions answered.
Short runbook	Steps to run the project, refresh data, rebuild Gold outputs and refresh/report in Power BI.
Git repository evidence	Repository with project files, meaningful commits, branch or PR history if used, and no secrets.
Final presentation	Short group presentation explaining the selected datasets, Medallion architecture, cleaning process, final model, quality checks and Power BI results.

Instructor review checkpoints

Checkpoint	What to show	Why it matters
1. Selected datasets and problem statement	Chosen datasets or shortlist, source links/files, join idea, problem statement, fallback plan.	Prevents teams from choosing one flat dataset or unrelated sources.

2. Product Backlog and team split	PBIs, owners/pairs, Definition of Done, and which Python cleaning work each member will do.	Ensures real work distribution and avoids one person doing all coding.
3. Medallion architecture plan	Source -> Bronze -> Silver -> Gold plan, layer purposes, output names and quality checks.	Prevents Bronze/Silver/Gold from becoming just folder names with no quality difference.
4. Final data model and grain	Gold model/mart, fact grain, dimensions/lookups, measures and Power BI source tables.	Prevents unclear row meaning and weak Power BI reporting model.
5. Python cleaning and validation sanity check	One or two examples of cleaned data, rejected/review rows, validation summaries and assigned contributors.	Confirms that Python cleaning is actually happening and not left to the end.
6. Power BI report check	Report connected to Gold outputs, pages/views, measures and business questions.	Ensures the final report is based on the engineered data model, not raw files.

Final presentation

Prepare a short presentation of the group result. The presentation should be practical and technical enough to show that the team understands the solution.

- Briefly describe the selected datasets, source types and problem statement.
- Show the Product Backlog or task board and explain the team split.
- Show the Medallion architecture and explain what Bronze, Silver and Gold mean in your project.
- Show examples of Python cleaning and validation work, including rejected/review rows or quality checks.
- Show the final Gold data model and explain the grain of the main fact/mart table.
- Show at least one validation query or one validation summary.
- Show the Power BI report and explain which business questions it answers.
- Mention limitations and what your team would improve next if there was more time.

Extra examples (if time permits)

- Add an incremental or date-based loading strategy.
- Add a run metadata table or file with run id, run timestamp, row counts and status.
- Create a separate rejected/quarantine output for bad source rows.
- Add a history-tracking idea for one changing dimension or changing source attribute.
- Add a second Gold mart for a different reporting question.
- Add more advanced Power BI measures or drill-through pages.
- Use SQL to create additional validation queries or Gold aggregations.

Common design mistakes to avoid

- Using only one flat CSV with no enrichment or modeling challenge.
- Choosing datasets that cannot be joined or compared in a meaningful way.
- Having no clear grain: the team cannot explain what one row means in the main table.
- Calling folders Bronze, Silver and Gold even though the data quality is the same in every layer.
- Doing all business calculations only inside Power BI without documenting the Gold model.
- Letting one person do all Python cleaning work.

- Building Power BI directly from raw source files instead of Gold outputs.

Project quality checklist

- At least two data sources are used, and at least one comes from the web/open data/API/download.
- The sources are realistic, approved and do not contain sensitive personal data.
- The team has a Product Backlog with owners or responsible pairs.
- The team can show meaningful Git usage and project history.
- Every team member participated in Python cleaning, ingestion, validation or transformation work.
- Bronze, Silver and Gold have clearly different purposes and data quality levels.
- The Silver layer contains cleaned, typed and standardized data.
- There are meaningful data quality checks.
- The Gold model has a clear grain and is suitable for analysis.
- Power BI uses Gold outputs as its reporting source.
- The project has a short runbook and can be demonstrated end-to-end or with clearly documented manual steps.
- The final presentation explains the datasets, architecture, cleaning, validation, Gold model and report results.